

Erdős Problem #319: progress report and formalization roadmap

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Executive summary

This is not a claimed solution of Erdős Problem #319. The current public status is still open. What has been achieved in the past few days is a substantial reframing and a set of proof-level intermediate targets:

1. A clean signed-circuit formulation of the problem.
2. A spectrum-intersection criterion for primitivity.
3. A separation between the known Croot/Adenwalla lower-bound mechanism and the genuinely open upper-bound/minimality problem.
4. Finite computational evidence near the constant $1 - 1/e$, useful as validation but not as proof.
5. A singleton-side theorem candidate / subclass result, subject to proof audit: the Croot/Adenwalla constant is expected to be asymptotically optimal when one sign class has exactly one element.
6. A fixed-Q proof programme using dynamic gates, target selection, and Liu-Sawhney-type reachability.
7. A Lean-facing scaffold in which the elementary combinatorial layer can be verified, while the analytic unit-fraction machinery is isolated as explicit black-box assumptions.

The most honest subsidy framing is: we have converted a difficult open Erdős problem into a structured formalization and proof-search programme with genuine intermediate deliverables. We have not solved the full problem.

1. Statement and notation

Let

$$R(X) = \sum_{x \in X} 1/x.$$

For a signed set A subset $\{1, \dots, N\}$, write

$$A = P \text{ disjoint_union } Q,$$

where P is the positive side and Q is the negative side. The signed zero-sum condition is

$$R(P) = R(Q) = S.$$

Erdős #319 asks for the largest size of such an A subject to the additional minimality/primitivity condition: no non-empty proper signed subcollection has total signed reciprocal sum zero.

Define the subset-sum spectra

$$\text{Sigma}(P) = \{ R(P') : P' \text{ subset } P \},$$

$$\text{Sigma}(Q) = \{ R(Q') : Q' \text{ subset } Q \}.$$

Then A is primitive exactly when

$$\text{Sigma}(P) \text{ intersection } \text{Sigma}(Q) = \{0, S\}.$$

This is one of the central reductions. It turns #319 into a two-colour reciprocal subset-sum spectrum intersection problem.

2. Known lower bound

Croot's short-interval unit-fraction theorem implies that for rational $r > 0$ one can represent r as a sum of reciprocals with denominators in a short interval. For $r = 1$, this gives a set B in approximately $[(1/e)N, N]$ with

$$\sum_{b \in B} 1/b = 1.$$

Then

$A = \{1\} \cup B$

with signs -1 on 1 and +1 on B gives a signed reciprocal zero-sum circuit. Since all reciprocals in B are positive, no proper subset of B can still sum to 1 after deleting a term. This gives the known lower bound

$$c(N) \geq (1 - 1/e + o(1)) N.$$

This is lower-bound progress only. It does not prove that larger primitive signed circuits are impossible.

3. Computational evidence

The exact-rational and Croot-style interval searches found near-misses around $N = 50$ and $N = 60$. For example, the $N = 60$, $\alpha = 0.340$ search produced a support of size 37 with reciprocal sum extremely close to 1. If repaired to exact equality, this would give $|A| = 38$ and $38/60 = 0.6333\dots$, close to

$$1 - 1/e = 0.6321\dots$$

This supports the Croot/Adenwalla constant numerically. It is not a proof, because #319 requires exact equality and minimality, and finite examples do not prove an asymptotic upper bound.

4. Proof-level output A: spectrum-intersection criterion

The following is an elementary theorem and should be the first Lean-verifiable lemma.

Theorem. Let P and Q be disjoint finite sets of positive integers with $R(P) = R(Q) = S$. Then the signed circuit $P \cup Q$ is primitive if and only if

$$\text{Sigma}(P) \cap \text{Sigma}(Q) = \{0, S\}.$$

Proof. A signed subcircuit is determined by subsets $P' \subseteq P$ and $Q' \subseteq Q$. Its signed reciprocal sum is zero exactly when $R(P') = R(Q')$. The empty-empty choice gives 0, and the full-full choice gives S . Any other equality gives a non-empty proper zero-subcircuit. Conversely, any non-empty proper zero-subcircuit gives such an equality. Therefore primitivity is exactly the assertion that the only spectral collisions are 0 and S .

This proof is fully formalizable without analytic number theory.

5. Proof-level output B: singleton-side optimality candidate

Define $c_1(N)$ to be the largest size of a primitive #319 circuit in $\{1, \dots, N\}$ in which one sign class has exactly one denominator.

Theorem. As N tends to infinity,

$$c_1(N) = (1 - 1/e + o(1)) N.$$

Proof sketch. If the singleton side is $\{d\}$, then the other side Q satisfies

$$R(Q) = 1/d.$$

For a fixed number k of denominators in $\{1, \dots, N\}$, the smallest possible reciprocal sum is obtained by taking the k largest denominators. Thus, if $k = \alpha N$, the minimum possible reciprocal mass is asymptotically

$$\int_{1-\alpha}^1 \frac{dx}{x} = \log(1/(1-\alpha)).$$

Therefore the condition $R(Q) = 1/d$ implies

$$\log(1/(1-\alpha)) \leq 1/d,$$

so

$$\alpha \leq 1 - \exp(-1/d) + o(1).$$

This is maximized at $d = 1$, giving $\alpha \leq 1 - 1/e + o(1)$. The matching lower bound is the Croot/Adenwalla construction with singleton denominator 1.

This is not a full solution of #319, because in the full problem both sign classes may be large. It should be treated as a theorem candidate / subclass result until the proof is audited and written independently.

6. Proof-level output C: large-prime-factor obstruction

The working lemma is a p-adic top-layer obstruction: very large minimal signed reciprocal circuits cannot freely use denominators with very large prime factors. In the current notes it is formulated roughly as follows.

Lemma candidate. If A is a primitive signed reciprocal circuit, $|A| > K$, and K is chosen so that the p-adic top-layer argument applies, then every denominator n in A has largest prime factor at most about N/K . With K around $\log N$, this gives a saving of order

$$N \log \log N / \log N$$

from the trivial upper bound N .

Status: mathematically plausible and useful as a structural obstruction, but modest. It should be treated as a warm-up lemma, not as a breakthrough. Its exact hypotheses should be rechecked before being presented as a final theorem.

7. Fixed-Q programme

Fix a finite negative side Q , and let

$$S_Q = R(Q).$$

The upper-bound task becomes: if $P \subseteq \{1, \dots, N\}$ satisfies

$$R(P) = S_Q$$

and

$$|P| \geq (\theta + \epsilon) N,$$

where $\theta = 1 - 1/e$, must P contain a subset B with

$$R(B) = R(Q')$$

for some non-empty proper Q' subset Q ? If yes, then $B \cup Q'$ is a proper signed zero-subcircuit, contradicting primitivity.

This is the fixed- Q target-hitting problem.

Dynamic gate

For a prime power q , define a tail-avoidance reciprocal-mass threshold $\Phi_q(u)$: the asymptotic minimum reciprocal mass of uN integers in $\{1, \dots, N\}$ when multiples of q are forbidden. The dynamic gate principle is:

$$\text{if } S_Q < \Phi_q(\theta + \epsilon),$$

then P must contain quantitatively enough q -divisible mass.

This converts the balance equation $R(P) = S_Q$ into local divisibility forcing.

Conditional fixed- Q theorem candidate

For fixed Q and $\epsilon > 0$, suppose there is a non-empty proper target Q' subset Q such that every prime power appearing in the denominator of $R(Q')$ is ϵ -good, i.e. controlled by the dynamic gate. Assuming the relevant Liu-Sawhney target-reachability interface, no primitive circuit $P \cap Q = \emptyset$ can have

$$|P| \geq (\theta + \epsilon) N$$

for all large N .

This is a serious partial theorem candidate. The remaining analytic black box is a target- r version of the Croot/Bloom/Liu-Sawhney unit-fraction machinery.

8. Arbitrary fixed Q versus growing Q_N

For arbitrary fixed Q , the project is now much sharper than before. The hard part is no longer vague; it is the small- ϵ regime and target selection. A possible route is:

1. For each fixed Q , list all proper targets Q' subset Q .
2. Compute the excess threshold $\epsilon_{Q'}^{Q'}$ needed for their denominator prime powers.
3. If $\epsilon > \epsilon_{Q'}^{Q'}$, use dynamic gate plus Liu-Sawhney reachability.
4. If $\epsilon \leq \epsilon_{Q'}^{Q'}$, prove a near-extremal stability or gate-absorption lemma.

For growing Q_N , the problem is harder. Constants depending on Q may collapse as N grows, and the pruning/reachability machinery may lose the very layers needed for the target. Full #319 likely requires a two-sided spectrum theorem rather than a fixed-target theorem.

9. New target-selection idea: cross-number route

The newest idea in the notes is a cross-number target-selection argument. It attempts to show that if $S_Q > 1$, then some proper target Q' subset Q has denominator supported only on the forced prime set. If correct, this would greatly strengthen the fixed- Q programme.

However, this should currently be written as a candidate lemma, not as a proved theorem, until the following implication is audited:

target-selection fails $\Rightarrow$ the associated sequence is a minimal zero-sum atom.

That implication is where hidden assumptions may enter. The cross-number inequality itself is plausible, but the reduction to an atom needs verification.

10. Lean/Aristotle formalization status

The Lean-facing decomposition should be:

Layer 1: elementary definitions, verifiable now

- reciprocal sum $R(X)$;
- signed circuit P disjoint_union Q ;
- primitive/minimal circuit;
- subset-sum spectrum $\Sigma(X)$;
- proof that primitivity is equivalent to $\Sigma(P) \cap \Sigma(Q) = \{0, S\}$.

Layer 2: elementary finite lemmas, formalizable next

- singleton-side upper bound, modulo standard harmonic-sum estimates;
- large-prime-factor obstruction, after exact p -adic hypotheses are finalized;
- finite Fourier subset-sum bridge in finite abelian groups.

Layer 3: analytic black boxes

- Croot short-interval unit-fraction theorem;

- Bloom density theorem;
- Liu-Sawhney target-reachability theorem, especially target-r rather than only target 1;
- smooth/pruning/local-mass lemmas.

A good Aristotle/Lean claim is therefore:

We have separated the proof into a Lean-verifiable combinatorial core and explicit analytic proof obligations. The project has not yet produced a completed Lean proof of Erdős #319.

11. Honest status table

Item	Status
Full Erdős #319	Open. Not solved.
Full Lean proof	Not available.
Spectrum-intersection reformulation	Proven/elementary; ready for Lean.
Croot/Adenwalla lower bound	Known literature; explains $1 - 1/e$.
Computations near $N = 50,60$	Useful evidence; not proof.
Singleton-side theorem	Candidate subclass result; proof audit needed.
Large-prime-factor obstruction	Useful but modest; audit exact statement.
Fixed-Q dynamic-gate theorem	Serious conditional theorem candidate.
Cross-number target selection	Promising but unaudited candidate.
Growing Q_N / full #319 upper bound	Long-term; needs new two-sided spectrum or stability theorem.

12. Suggested subsidy wording

Over the first few days, we did not solve Erdős Problem #319, but we achieved a structured reduction and formalization programme. We reformulated the problem as one about primitive signed reciprocal circuits and subset-sum spectrum intersections. This separated the known Croot/Adenwalla lower-bound construction from the genuinely open upper-bound/minimality direction.

We isolated a singleton-side subclass result candidate: the known constant $1 - 1/e$ should be asymptotically optimal among singleton-side signed circuits, subject to a proof audit. We also identified a large-prime-factor obstruction and a fixed-Q dynamic-gate strategy that reduces several cases to explicit target-reachability inputs from Croot/Bloom/Liu-Sawhney-type unit-fraction machinery.

The current research frontier is the arbitrary fixed-Q problem. It has been reduced to target selection, small-epsilon near-extremal stability, and a target-r Liu-Sawhney interface. The full problem, where both sign classes may grow with N , remains substantially harder and likely requires a two-sided spectrum-intersection theorem.

This is genuine early-stage progress: not a completed solution, but a clear programme with formal definitions, proof obligations, computational tests, and at least one clean theorem suitable for a short internal note.